Agentic AI-Based Multi-Agent Recommender System for Higher Education Decision Support

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***Abstract*—Students choosing a university and program must weigh academic fit, career goals, financial constraints, location, and employability, usually with no prior interaction history to learn from. Classical educational recommenders address this pre-admission, cross-institution decision through collaborative, content-based, or knowledge-based filtering but remain static and weakly explainable. Recent agentic, LLM-powered recommenders add adaptivity, planning, memory, and explainability, but published 2025–2026 systems target already-enrolled students in an already-chosen program, not the pre-admission choice. We propose a five-agent architecture (profiling, retrieval, multi-objective recommendation, explanation, and memory-reflection) for this gap, evaluated against re-implemented classical baselines (HRSPCA, StudieMe) and a single-prompt LLM ablation on a released synthetic benchmark. The architecture does not win on oracle-graded top-k accuracy, but wins decisively on explanation quality: 91–100% of explanations are grounded in retrieved evidence against 36–67% for the single-prompt baseline. Memory-and-reflection improves session-over-session quality. Explainability, not accuracy, is the architecture's demonstrated contribution; the prototype and dataset are open-sourced.**

***Index Terms*—agentic AI, multi-agent systems, recommender systems, higher education, large language models, explainable recommendation, decision support**

I. Introduction

Choosing a university and program is a high-stakes, low-frequency decision made under incomplete information. Prospective students must reconcile academic interests, career goals, skills, financial constraints, location preferences, and employability outcomes, usually without any prior interaction history a recommender system could exploit: the canonical cold-start setting. A substantial body of work addresses this decision with collaborative, content-based, knowledge-based, and hybrid or machine-learning classification approaches [1]–[6]. These systems established strong evaluation methodology and real datasets, but a 2024 systematic review documents recurring limitations: static pipelines, minimal conversational interaction, and weak explainability [7].

Separately, LLM-powered agents have produced a new recommendation paradigm: systems that plan, retrieve, reflect, and maintain long-term memory rather than scoring a fixed candidate set once [8]. Agentic recommenders of this kind have been demonstrated in e-commerce and general content domains [9]–[11], and, more recently, in education, but specifically for students who are *already enrolled* in an *already-chosen* program: advising Q&A [12], learning-pathway sequencing [13], and course selection [14]. None of these systems address the earlier, arguably higher-stakes decision of *which* institution and program to choose.

This paper's central claim is that this earlier decision (pre-admission, cross-institution selection) is a distinct decision scope from the in-program advising/pathway/course-selection problems 2025–2026 agentic education systems solve, and remains addressed only by pre-agentic methods.

Contributions: (i) an agentic multi-agent architecture targeting pre-admission, cross-institution decision support, distinct from in-program agentic-education work (Section II); (ii) an explicit multi-objective reasoning mechanism evaluated against classical baselines on both standard IR metrics and agentic-specific metrics (explanation quality, adaptivity, cold-start robustness) that recent surveys call for but no education-specific evaluation yet supplies [8]; (iii) a working prototype, evaluation harness, and results, released as an open-source artifact: https://github.com/samuelshine/multi-agent-education-recommender-system.

II. Related Work

HRSPCA [1] and StudieMe [2] target the same pre-admission, cross-institution scope as this paper, via hybrid collaborative filtering with knowledge-based rules [1] or content-based TF-IDF similarity [2]. Somyürek and Aksoy [3] extend this to joint university-and-major selection, the closest classical match to the decision scope adopted here. Elahi et al. [5] evaluate a collaborative-filtering university recommender via offline cross-validation plus an online user study, a protocol this paper adopts as a template (Section V). El-Keiey et al. [15] and Hill et al. [6] (both 2025) address adjacent career/program recommendation with single-model, non-agentic pipelines. None reason via autonomous agents, maintain cross-session memory, or generate grounded explanations, confirmed as recurring limitations by a systematic review [7].

In the general-domain agentic literature, Portugal et al. [9] propose a multi-agent recommender framework whose title closely resembles this paper's title (the closest general-domain precedent) but target e-commerce with no educational features or metrics. Narasimhan and Narasimhan [10] introduce

TABLE I
POSITIONING AGAINST THE CLOSEST PRIOR WORK

System	Target user	Decision scope	Method
[1], [2], [3]	Prospective	Pre-admission, cross-institution	CF / CB / KB (pre-agentic)
[12]	Enrolled	In-program advising	Multi-agent, human-in-loop
[13]	Enrolled	In-program pathway	Multi-agent (18+ agents)
[14]	Enrolled	In-curriculum course	Multi-agent + KG (3 agents)
This work	**Prospective**	**Pre-admission, cross-institution**	**Multi-agent LLM (5 agents)**

a general-domain agentic benchmark; no education-specific one yet exists. Peng et al.'s survey [8] covers e-commerce, media, and conversation; higher education appears in neither its applications nor its future directions.

Multi-agent educational recommenders predate LLMs; the novelty here rests on combining agentic LLM capabilities (planning, memory, reflection, grounded explanation) with the pre-admission scope, not on multi-agent structure alone. Three 2025–2026 systems combine multi-agent LLM methods with an educational target. AdvisingWise [12] is a human-in-the-loop system answering advising questions for *already-enrolled* students, with every response human-validated; it does not recommend a program. Chaturvedi and Gunawardena [13] deploy an 18-agent hierarchy for in-program pathway recommendation at production scale, addressing only post-enrollment adaptation. Zhong et al. [14] propose a three-agent architecture (profiling, retrieval, generation) with a course knowledge graph (structurally the closest precedent to Section III), but scoped to in-curriculum course choice, without multi-objective reasoning over career, financial, location, and employability constraints. Table I summarizes the resulting gap.

III. Proposed Architecture

An initial seven-agent brainstorm (profiling, goal planning, retrieval, recommendation, reflection, explanation, memory) was merged to five: Zhong et al. [14] report gains with only three agents, while Chaturvedi and Gunawardena's 18+ agents [13] require production-scale engineering not reproducible in a small-team research prototype. Explanation and memory-reflection remain separate, visible components since they map directly to this paper's headline claims; goal-planning's constraint logic folds into profiling and recommendation.

Profiling & Constraint Agent extracts a structured student-constraint object single-turn per session; continuity comes from the Memory & Reflection Agent's persisted state (Fig. 1, dashed arrow). **Retrieval Agent** selects programs worth presenting from a deterministic hard-constraint prefilter's output (admission, budget, location); this is LLM-based relevance selection over a prefiltered candidate list, not embedding retrieval, following [11]. **Recommendation & Planning Agent** ranks candidates under multiple, potentially conflicting objectives: the mechanism RQ2 evaluates. **Memory & Reflection Agent** reviews the shortlist for constraint violations and persists state across sessions: the mechanism RQ3 evaluates; a second, deterministic guard repeats the hard-constraint check after it. **Explanation Agent** generates a per-recommendation rationale grounded in the Retrieval Agent's evidence, addressing the explainability limitation [7] documents in classical systems. Relative to the closest agentic precedent's profiling, retrieval, and generation agents [14], the new elements are constraint elicitation over financial, location, and employability objectives (Profiling), explicit trade-off ranking across them (Recommendation), a reflection pass with cross-session memory (Memory & Reflection), and an explanation step restricted to retrieved evidence (Explanation). The two deterministic prefilter/guard components, not agent judgment alone, actually guarantee hard-constraint satisfaction. The 7→5 merge is a buildability judgment, not an ablation result; H3's memory-and-reflection results (Section VI, evaluated as a single fused mechanism) provide the retrospective check and motivate no split back apart, though a finer-grained ablation isolating the two is left to future work.

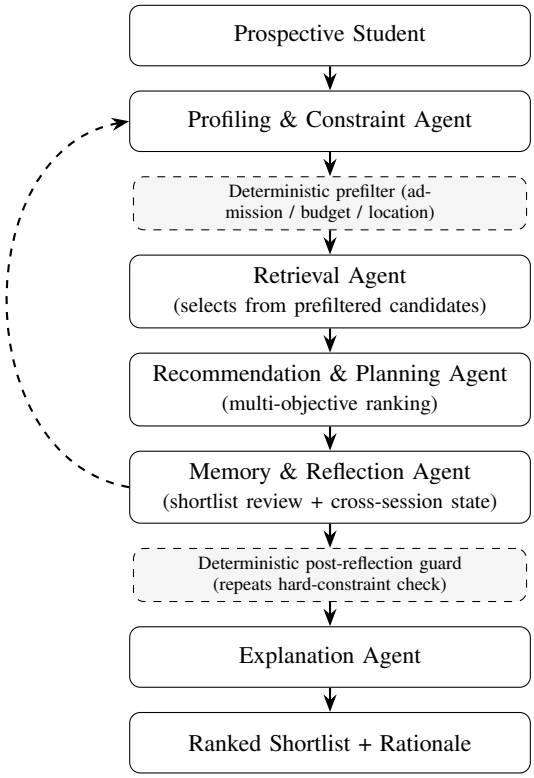


Fig. 1. Proposed five-agent architecture and pipeline flow.

IV. Research Design

RQ1. Does the architecture improve explainability and adaptivity over classical recommenders [1]–[3] within a 5% non-inferiority margin of their accuracy (precision@k, recall@k, NDCG)?

RQ2. How should multi-objective constraints be represented and reasoned over by specialized agents, compared to a single monolithic LLM prompt or classical hybrid scoring?

RQ3. Do the Memory & Reflection Agent's mechanisms measurably reduce cold-start degradation compared to a stateless pipeline?

H1. The architecture achieves higher explanation quality than a classical hybrid baseline re-implemented from [1], [2], within a pre-registered[1] 5% relative non-inferiority margin on NDCG@k.

H2. Explicit multi-objective reasoning improves relevance for students with conflicting constraints relative to a single-objective baseline.

H3. Memory & Reflection mitigates cold-start degradation more than a stateless pipeline, across simulated first-vs.-later-session interactions.

Datasets. The intended primary H1 benchmark is Elahi et al.'s [5] university-rating dataset (1,515 ratings, 80 users, 551 universities), available on request; that request was still pending when experiments ran, so results here use a released synthetic constraint-preference benchmark instead (a self-constructed-benchmark limitation, Section VII), which also carries the H2/H3 conditions rating-only data cannot express. It regenerates deterministically from 300 synthetic students sampled from the public 20,000-record IT-student corpus of [15] (financial, location, and employability constraints attached by documented rules), a fictional 200-program catalog, and graded relevance (0–3) from a published oracle utility function over academic fit, financial feasibility, location, and employability, fixed before evaluation. Each student carries a three-session progressive-disclosure script; 162/300 students have a financially infeasible best-fitting program (the H2 condition).

Baselines. (a) a re-implementation of HRSPCA's cascaded rule/predictor pipeline [1]; (b) a re-implementation of StudieMe's content-based ranker [2]; (c) a single-prompt LLM baseline performing profiling, ranking, and explanation in one call, following [14]. Both classical baselines were re-implemented from the original papers' full texts, with every underspecified detail resolved by a logged best-faith choice (13 choices, released in the artifact); two original components could not be expressed in the evaluation catalog, favoring the baselines (Section VII). All agentic conditions run on a local open-weights Gemma model via Ollama (`gemma4:e4b`), provider-agnostic in the released artifact.

[1]Metric choice, $k = 5$, and the 5% margin were committed to the public repository (commit `9b47ce4`, 2026-07-17) before any agentic-condition result was computed; results were written up in a later commit (`ba7757e`, 2026-07-20).

TABLE II
Accuracy on the held-out evaluation split ($n = 100$, full information; zero pipeline crashes, 4 validated-fallback events across 500 agent calls). Bold marks the column maximum among performance metrics.

System	NDCG@5	P@5	R@5	|list|
HRSPCA (re-impl. [1])	0.793	0.796	0.061	5.0
StudieMe (re-impl. [2])	0.673	0.760	0.113	5.0
Single-prompt baseline	0.830	0.912	**0.166**	4.9
Single-prompt (grounded)	**0.831**	**0.926**	0.159	4.8
Five-agent v1 (pre-registered)	0.706	0.904	0.129	3.9
Five-agent v2 (revised)	0.806	0.898	0.156	4.8

v1: pre-registered pruning policy, abstains rather than pads a shortlist below k. v2: post-hoc revision backfilling to k, dropping only for hard-constraint violations. Grounded: single-prompt baseline with Explain's grounding instruction added.

V. Evaluation Plan

The headline metric is NDCG@5 with graded relevance (0–3), chosen for rank-sensitivity and because it uses the oracle's graded relevance directly rather than a binary cutoff; $k = 5$ matches both the baselines' reconstructed output length and a plausible shortlist size. Precision@5 and recall@5 are secondary; all systems are evaluated on the same held-out split of 100 students, under the pre-registered 5% non-inferiority margin (H1).

Explanation quality is scored by LLM-as-judge on a three-dimension rubric (clarity, groundedness, usefulness; 1–5 each), following [12]. To avoid self-preference bias, the judge is a different model family than the experiment model (primary: Qwen-family 14B-class; a Mistral-family second judge gives inter-judge agreement), with system identity blinded and absolute scoring.

Conditions: (i) the full five-agent pipeline; (ii) a no-memory ablation (H3); (iii) a single-objective ablation (H2); (iv) the single-prompt baseline; (v) the two classical baselines. H1 uses the full-information state; H3 uses the three-session progressive-disclosure structure, with simulated-user feedback derived from the oracle's utility ranking, identical across every system; this is discussed as a limitation in Section VII.

VI. Results

A. Top-k accuracy (H1)

Table II reports accuracy on the held-out 100-student split. HRSPCA and StudieMe are the published state of the art for this exact decision scope (Table I); the agentic-education systems [12]–[14] solve different tasks for already-enrolled students, so they cannot be run as baselines on a pre-admission benchmark. The strongest classical baseline, HRSPCA, reaches NDCG@5 $= 0.793$, so the pre-registered 5% margin requires NDCG@5 ≥ 0.753.

The pre-registered accuracy criterion is not met. Five-agent v1 reaches NDCG@5 $= 0.706$, an 11.0% relative shortfall against HRSPCA, and the single-prompt baseline (0.830) outperforms both. **Diagnostics** attribute this to shortlist *abstention*, not ranking quality: Reflection trims candidates

rather than padding with constraint-violating ones, so 70% of shortlists contain fewer than five items (mean 3.9), and NDCG@5 penalizes every unfilled position. Measured at emitted length, v1 scores 0.827 against 0.830 (paired bootstrap, $n = 100$: mean diff -0.005, 95% CI $[-0.024, +0.014]$, not significant), and precision@5 matches. A post-registration revision (v2, reported alongside not in place of v1) restores list length to 4.8 by dropping only for hard-constraint violations; it reaches NDCG@5 $= 0.806$, clearing the pre-registered bar and slightly exceeding HRSPCA, confirming the abstention diagnosis, but remains 2.9% relative below the single-prompt baseline, so neither pruning policy shows a top-k accuracy advantage over a single LLM call.

B. Multi-objective conflict (H2)

On the 56 conflict-subset students, v1's multi-objective pipeline scores NDCG@5 $= 0.666$ against 0.718 single-objective, again driven by shorter conflict-case shortlists (mean 3.70 vs. 4.02). Under v2, multi-objective reaches 0.792 (mean 4.77 items) against 0.793 single-objective (paired bootstrap, $n = 56$: mean diff -0.001, 95% CI $[-0.017, +0.014]$, not significant). **H2 is not supported on oracle-graded accuracy**: the oracle's utility already encodes all four objectives, so a single-objective ranker over a constraint-prefiltered pool recovers most of the signal; whether multi-objective reasoning earns its place in *explanations* is assessed below.

C. Cold start and memory (H3)

On a fixed 30-student progressive-disclosure cohort, full and no-memory conditions are identical at session 1 by construction (both NDCG@5 $= 0.583$). With memory, session-2 accuracy reaches 0.776 vs. 0.735 without ($+0.041$), and the gap persists at session 3 (0.748 vs. 0.729); recall@5 degrades session-2-to-3 without memory ($0.126 \rightarrow 0.090$) but holds with it ($0.118 \rightarrow 0.115$). The effect is directionally consistent but modest at $n = 30$; progressive disclosure itself is the largest driver of improvement in both conditions. H3 is supported in direction, with the caveat that "cold-start mitigation" is more precisely an *adaptivity* gain: the first session is unaffected by construction.

D. Explanation quality (H1)

Table III reports cross-judge mean rubric scores. **H1's explanation-quality claim is supported**, by a wide margin on groundedness (4.92 v1/v2 vs. 3.96 single-prompt) and usefulness (3.65/3.60 vs. 2.80); clarity shows a smaller gap (4.11/4.06 vs. 3.69). The v1/v2 accuracy difference leaves explanation quality essentially unchanged, indicating the two properties are largely independent here.

Isolating architecture from instruction. The single-prompt baseline's explanation instruction is weaker than EXPLAIN's, confounding architecture with wording. A fourth condition, single-prompt (grounded), upgrades only the instruction to EXPLAIN's directive verbatim (Table II); it scores clarity 3.81, groundedness 4.47, usefulness 3.23 (between the ungrounded baseline and the five-agent conditions), and its NDCG@5/precision@5 match or slightly exceed the ungrounded baseline's, so the instruction change costs nothing on ranking. Decomposing the ungrounded-to-v1 gap (paired bootstrap, $n = 100$): the instruction effect accounts for 53% of the groundedness gap ($+0.515$, 95% CI $[+0.320, +0.705]$), 50% of usefulness ($+0.425$, 95% CI $[+0.295, +0.545]$), and 29% of clarity ($+0.120$, 95% CI $[+0.035, +0.205]$); the residual architecture effect accounts for the rest, and none of the six intervals crosses zero. Both grounding instructions and architectural decomposition make a real, statistically supported contribution, with neither alone accounting for it.

This is corroborated by an invented-claims failure signal: the single-prompt baseline was flagged in 64/100 explanations by the primary judge (33/100 by the agreement judge) against 5–9/100 for the five-agent conditions; the grounding instruction alone cuts the baseline's rate to 42/100 and 6/100. Inter-judge agreement is correspondingly tighter for the five-agent conditions (mean |diff| 0.09–0.15) than the single-prompt family (1.26). Restricting this analysis to the 56 H2 conflict students reproduces the gap unchanged: multi-objective reasoning does not win on oracle-graded ranking, but produces more reliably grounded explanations for students facing a real trade-off.

E. Component ablation summary

Table IV collects the four ablations against the full pipeline. Removing agent decomposition costs explanation groundedness but not accuracy; removing memory costs adaptivity; removing multi-objective ranking changes neither metric measurably. Retrieval, reflection, and the explanation agent were not ablated individually (Section VII).

TABLE III
EXPLANATION-QUALITY RUBRIC SCORES (1–5), CROSS-JUDGE MEAN OF `QWEN2.5:14B` AND `MISTRAL-NEMO:12B`, BLIND ABSOLUTE SCORING, $n = 100$. BOLD MARKS THE COLUMN MAXIMUM.

System	Clarity	Groundedness	Usefulness
HRSPCA (re-impl. [1])	—	—	—
StudieMe (re-impl. [2])	—	—	—
Single-prompt baseline	3.69	3.96	2.80
Single-prompt (grounded)	3.81	4.47	3.23
Five-agent v1 (pre-registered)	**4.11**	**4.92**	**3.65**
Five-agent v2 (revised)	4.06	**4.92**	3.60

HRSPCA/StudieMe are non-generative ranking pipelines with no natural-language output to score.

TABLE IV
COMPONENT ABLATIONS, ABLATED CONDITION VS. FULL FIVE-AGENT PIPELINE ON THE SAME SET (NDCG@5; GROUNDEDNESS 1–5, CROSS-JUDGE MEAN). H1/H2 ROWS USE V2; THE H3 COHORT WAS RUN ONLY WITH V1.

Ablation	Set	NDCG@5	Ground.
No memory (stateless)	H3, S2	0.735 / 0.776	—
Single-objective ranking	H2	0.793 / 0.792	—
Single prompt + grounding	H1	0.831 / 0.806	4.47 / 4.92
Single prompt	H1	0.830 / 0.806	3.96 / 4.92

VII. Discussion and Limitations

The gap argument in Section II was verified against full-text readings of the decisive prior work: all three agentic-education systems [12]–[14] address only already-enrolled students, and both re-implemented classical baselines [1], [2] contain no agentic, dialogue, or memory elements (peripheral citations remain verified at the abstract level only). The 7-to-5 agent merge is a buildability judgment, not an empirical result.

Oracle circularity and synthetic data. Ground-truth relevance is a published utility function fixed before evaluation, not human judgment; a system that reverse-engineered it would score perfectly. The oracle is documented in the released dataset card and identical for all systems, and explanation quality is judged independently of it. The catalog is fictional, constraints follow documented sampling rules, and the three-session structure is scripted with oracle-derived simulated feedback. Career coverage is IT-only, inherited from [15]; generalization beyond IT-bound students is untested.

Baseline reconstruction fidelity. Both baselines were re-implemented from their papers' full texts, with underspecified details resolved by logged best-faith choices (released in the artifact). Two original components (HRSPCA's college-capacity rule and StudieMe's faculty-matching stage) could not be expressed in the catalog; both omissions favor the baselines, not the proposed system.

LLM-as-judge validity and model scale. Explanation scores come from LLM judges, not human raters; cross-family judging, blinding, and two-judge agreement reduce but do not eliminate known judge biases, and this study performs no human-subject validation. Retrieval, reflection, and the explanation agent were not ablated separately, so their individual contributions to the groundedness gain are not isolated. All agentic conditions run on one local model (`gemma4:e4b`), so conclusions are conditioned on this model class; the evaluation split ($n = 100$) is larger than the closest classical baselines' user studies but small in absolute terms.

VIII. Conclusion and Future Work

For pre-admission program selection, agentic decomposition improves explanation quality, not ranking accuracy. Three conclusions follow from the evidence. First, accuracy and explainability are separable: the five-agent pipeline matches or exceeds the classical state of the art (NDCG@5 0.806 vs. 0.793 for HRSPCA) but not a single LLM call (0.830), while grounding 91–100% of its explanations in retrieved evidence against 36–67% for that call, with far fewer invented claims. Second, about half of this groundedness gain comes from instruction wording and half from the architecture's dedicated explanation step over agent-curated evidence; both effects are statistically supported. Designers who cannot deploy a multi-agent pipeline can therefore get part of the benefit from a grounding instruction alone. Third, multi-objective ranking adds no oracle-graded accuracy when hard constraints are already prefiltered, and cross-session memory gives a modest adaptivity gain ($+0.041$ NDCG@5 at session 2), not a cold-start fix. Agentic architectures for admission advising should thus be justified by trustworthy, auditable explanations rather than by ranking gains, at least at this model scale. These conclusions rest on synthetic data and LLM judges; the next steps are a human study with prospective students and academic advisors, evaluation on real preference data such as Elahi et al.'s [5], and a component-level ablation separating retrieval, reflection, memory persistence, and the explanation agent. The prototype, evaluation harness, and synthetic dataset are released as an open-source artifact so these directions do not depend on the authors' environment.

Data Availability Statement

The synthetic evaluation dataset, its generator, dataset card, and all run records and LLM-as-judge outputs underlying the reported results are released with the paper's open-source artifact at https://github.com/samuelshine/multi-agent-education-recommender-system under CC BY 4.0 (code under MIT). The public seed corpus of [15] is fetched from its original GPL-3.0 repository at generation time and not redistributed. The real-preference dataset of [5], requested for use as an H1 benchmark, had not been received at the time of writing.

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